Who Kang

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Education

University of Washington Seattle, WA
B.S. Aeronautics & Astronautics, Minor in Informatics | Cumulative GPA: 3.79 *Aug 2025 – Jun 2027*

FAA Airframe & Powerplant (A&P) Mechanic License | Federal Aviation Administration *Feb 2024*

Summary of Qualifications

- FAA-certified A&P mechanic and AeroAstro researcher combining flight-certified hardware expertise with research-grade embedded systems development.
- Designed and validated a SSVEP-based multimodal UAV control system achieving **98.8% command accuracy** across **270 trials** with zero misclassifications.
- Currently contributing to ML-assisted composite defect detection at UW G3MS Lab, with hands-on background in AFP layup, avionics integration, and composite airframe fabrication.

Experience

G3MS Lab, University of Washington Seattle, WA
Undergraduate Research Assistant *Mar 2026 – Present*

- Developing a machine-learning inspection pipeline in Python and MATLAB that automatically detects pre-cure AFP defects including overlaps, tow twisting, and misalignment from microscope images.
- Performed composite prepreg layup and coupon-level defect documentation; organized AFP coupon photo dataset to train classification and localization models for automated defect detection.
- Conducted 10+ Filabot extrusion trials with TotalEnergies Corbion LX175 bio-based PLA at 185°C, achieving consistent target diameter across 3 to 4 spools as a baseline process for future recycled and conductive filament development.

K-Capstone Challenge, Consulate General of the Republic of Korea in Seattle Seattle, WA
Industry Collaboration Researcher *Dec 2025 – Mar 2026*

- Finished **2nd place** evaluating U.S. market-entry feasibility for DOGU Robotics across 3 deployment environments.
- Developed a Robot Feasibility Index (RFI) scoring deployment scenarios across Need, Demand, Acceptance, Concern, and Gate factors; drove analytical shift from broad market research to a structured quantitative framework.
- Built a Python/pandas weighted-scoring pipeline for 113 survey responses and approximately 20 stakeholder interviews, producing a quantitative deployment ranking that supported the team's final market-entry recommendation.

Projects

Multimodal UAV Control Testbed | *Python, Raspberry Pi, ESP32, Betaflight* Jan 2026 – May 2026

- Built a custom UAV testbed for hands-free drone control combining voice recognition (Vosk) and SSVEP-based command verification through an ESP32 routing layer and Betaflight flight controller.
- Achieved **98.8% command accuracy** and **2.08 s mean latency** across **270 trials** with zero misclassifications; validated against voice-only (97.8%) and SSVEP-only (95.4%) baselines.
- Owned full UAV hardware stack including airframe assembly, propulsion sizing, DolphinRC F405 V3 Betaflight configuration, and ESP32 command pipeline; presented at UW Spring 2026 Undergraduate Research Symposium.

AIAA DBF-Style Aircraft Design & Build | *SolidWorks, Fusion 360, AVL, MATLAB* Aug 2024 – May 2025

- Founded and sized an 8-member team's aircraft: 6 ft wingspan, NACA 2411 airfoil, 7.5 aspect ratio, 611 in^2 wing area at Re $\approx 1.04 \times 10^5$; converged on rib-and-spar structure with 8 PLA ribs on carbon fiber spars.
- Applied AVL neutral point analysis to set CG target at 25% MAC; SimScale joint analysis identified epoxy-only vertical stabilizer bond as failure mode from prototype testing.
- Led team fabrication planning including wing rib cutting, carbon fiber spar bonding, and control surface linkage routing from concept through early prototype build.

IREC-Oriented High-Power Rocket | *OpenRocket, MATLAB* Apr 2024 – Jun 2024

- Modeled CG/CP stability in OpenRocket; upgraded from K-class to Aerotech M1350W-0 when simulation showed insufficient thrust for 8.8 lb payload to 10,000 ft.
- Simulation predicted **2,443 m apogee**, **Mach 0.723** max velocity, and **2.24-caliber** static stability margin; fabricated carbon fiber/fiberglass airframe with WEST SYSTEM epoxy and launched at OROC site, Brothers, OR.
- Rigged dual-deploy recovery system with drogue and main parachute; conducted pre-flight CG/CP verification and post-flight airframe inspection for full vehicle recovery.

Technical Skills

Programming / Analysis / Tool: Python, C/C++, MATLAB/Simulink, pandas, OpenRocket, Git, Microsoft Office
Embedded / Flight Hardware: ESP32, Raspberry Pi, Arduino, Betaflight, UART, IMU/GPS/Altimeter integration
CAD / Simulation: SolidWorks, Fusion 360, Onshape, AVL, SimScale
Manufacturing: Composite Layup, Automated Fiber Placement (AFP), FDM 3D Printing, Sheet Metal, Avionics Wiring